

Grand Award Judging Criteria

The following evaluation criteria are used for judging at the ISEF. As shown below, science and engineering have different criteria, each with five sections as well as suggested scoring for each section. Each section includes key items to consider for evaluation both before and after the interview.

Students are encouraged to design their posters in a clear and informative manner to allow pre-interview evaluation and to enable the interview to become an in-depth discussion. Judges should examine the student notebook and, if present, any special forms such as Form 1C (Regulated Research Institutional/ Industrial Setting) and Form 7 (Continuation/ Research Progression Projects). Considerable emphasis is placed on two areas: Creativity and Presentation, especially in the Interview section, and are discussed in more detail below.

Creativity: A creative project demonstrates imagination and inventiveness. Such projects often offer different perspectives that open up new possibilities or new alternatives. Judges should place emphasis on research outcomes in evaluating creativity.

Presentation/Interview: The interview provides the opportunity to interact with the finalists and evaluate their understanding of the project's basic science, interpretation and limitations of the results and conclusions.

- ☐ If the project was done at a research or industrial facility, the judge should determine the degree of independence of the finalist in conducting the project, which is documented on Form 1C (Regulated Research Institutional/ Industrial Setting).
- ☐ If the project was completed at home or in a school laboratory, the judge should determine if the finalist received any mentoring or professional guidance.
- ☐ If the project is a multi-year effort, the interview should focus ONLY on the current year's work. Judges should review the project's abstract and Form 7 (Continuation/ Research Progression Projects) to clarify what progress was completed this year.
- ☐ Please note that both team and individual projects are judged together, and projects should be judged only on the basis of their quality. However, all team members should demonstrate significant contributions to and an understanding of the project.

Judging Criteria for Science Projects

- I. **Research Question** (10 pts)
 - ☐ clear and focused purpose
 - ☐ identifies contribution to field of study
 - ☐ testable using scientific methods
- II. **Design and Methodology** (15 pts)
 - ☐ well-designed plan and data collection methods
 - ☐ variables and controls defined, appropriate and complete
- III. **Execution: Data Collection, Analysis and Interpretation** (20 pts)
 - ☐ systematic data collection and analysis
 - ☐ reproducibility of results
 - ☐ appropriate application of mathematical and statistical methods
 - ☐ sufficient data collected to support interpretation and conclusions
- IV. **Creativity** (20 pts)
 - ☐ project demonstrates significant creativity in one or more of the above criteria
- V. **Presentation** (35 pts)
 - a. **Poster (10 pts)**
 - ☐ logical organization of material
 - ☐ clarity of graphics and legends
 - ☐ supporting documentation displayed
 - b. **Interview (25 pts)**
 - ☐ clear, concise, thoughtful responses to questions
 - ☐ understanding of basic science relevant to project
 - ☐ understanding interpretation and limitations of results and conclusions
 - ☐ degree of independence in conducting project
 - ☐ recognition of potential impact in science, society and/or economics
 - ☐ quality of ideas for further research
 - ☐ for team projects, contributions to and understanding of project by all members

Judging Criteria for Engineering Projects

- I. **Research Problem** (10 pts)
 - ☐ description of a practical need or problem to be solved
 - ☐ definition of criteria for proposed solution
 - ☐ explanation of constraints
- II. **Design and Methodology** (15 pts)
 - ☐ exploration of alternatives to answer need or problem
 - ☐ identification of a solution
 - ☐ development of a prototype/model
- III. **Execution: Construction and Testing** (20 pts)
 - ☐ prototype demonstrates intended design
 - ☐ prototype has been tested in multiple conditions/trials
 - ☐ prototype demonstrates engineering skill and completeness
- IV. **Creativity** (20 pts)
 - ☐ project demonstrates significant creativity in one or more of the above criteria
- V. **Presentation** (35 pts)
 - a. **Poster (10 pts)**
 - ☐ logical organization of material
 - ☐ clarity of graphics and legends
 - ☐ supporting documentation displayed
 - b. **Interview (25 pts)**
 - ☐ clear, concise, thoughtful responses to questions
 - ☐ understanding of basic science relevant to project
 - ☐ understanding interpretation and limitations of results and conclusions
 - ☐ degree of independence in conducting project
 - ☐ recognition of potential impact in science, society and/or economics
 - ☐ quality of ideas for further research
 - ☐ for team projects, contributions to and understanding of project by all members